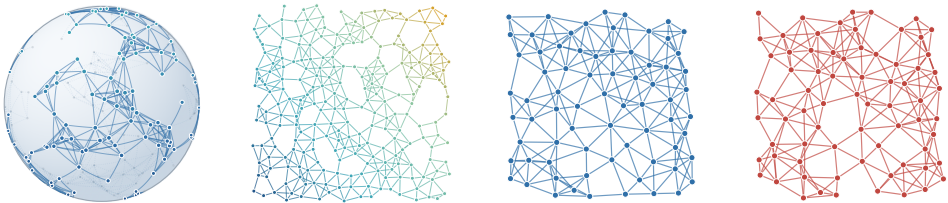
Matching Correlated Random Geometric Graphs

Shuangping Li
Yale Statistics & Data Science

Joint work with Dong Huang (Tsinghua University), Ankang Jiao (Peking University), and Zhangsong Li (Peking University)



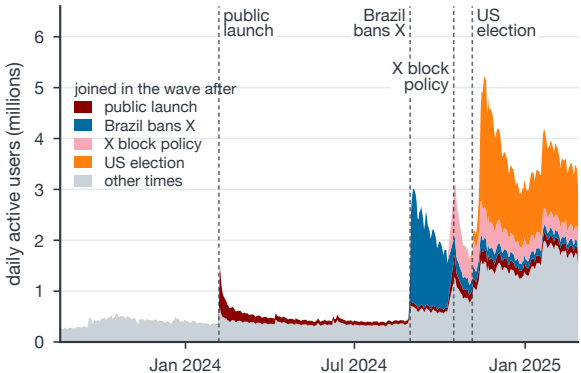
Probabilistic Phenomena on Graphs, Groups, and Manifolds
Simons Institute

Graph Matching in Random Geometric Graphs **I**: Efficient Algorithms, arXiv:2610.16675 · **II**: Information-Theoretic Limits

Outline

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2. Correlated random geometric graphs and our result
3. The algorithm: counting seeded paths
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5. Lower bounds and open problems

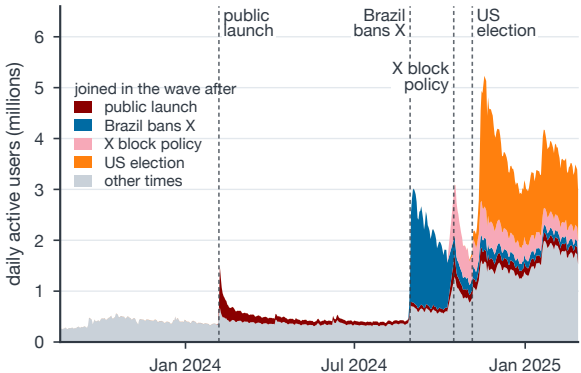
Motivating example: from X to Bluesky



Daily active users on Bluesky; the colors show when the users joined.
Data: Seckin et al. 2025 (CC BY 4.0).

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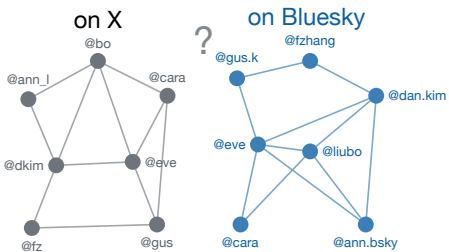
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- 276,431 academics on X, followed through 2023–24 Quelle Denker Garg Bovet 25:
  - 18% moved to Bluesky;
  - a contact who moved the day before more than doubles one's daily chance of moving;
  - those who quickly reconnect with their former contacts stay longer.

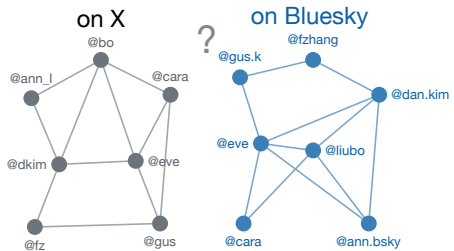
Linking the two networks



The same people, with different handles. Most follows are rebuilt.

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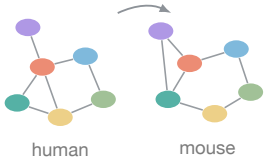
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- Social science: who moved, did communities move together, which ties were rebuilt? This needs a link from each person's X account to their Bluesky account Quelle et al. 25
- Names are unreliable: only 53% keep the same handle. But people rebuild their networks, so the two follow graphs are correlated graphs on (mostly) the same people. Link them by graph matching.

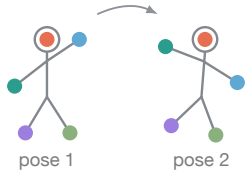
Applications



Computational biology

which proteins of two species correspond?

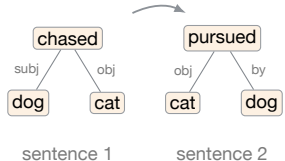
Singh Xu Berger 08; Kazemi et al. 16



Computer vision

which points of two scans correspond?

Lähner et al. 16

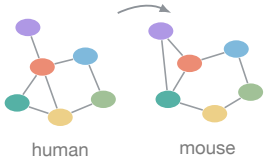


Natural language

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Haghighi Ng Manning 05

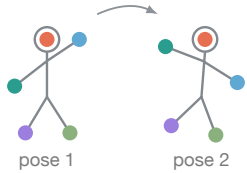
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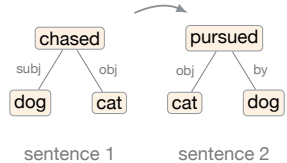
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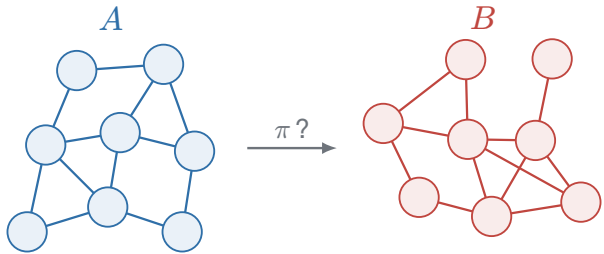
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- Two networks describe related systems, but there are no common labels (different species, different scans, different words).
- What is shared is the pattern of connections, and it may reveal the correspondence.

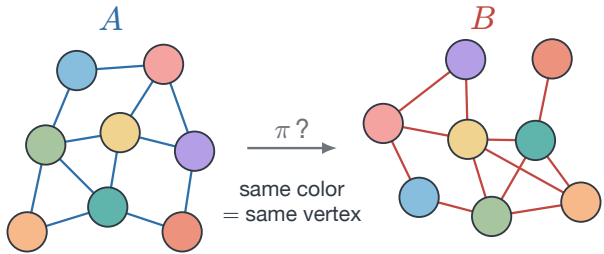
Graph matching

Given two graphs on the same n vertices, find the hidden correspondence between them.



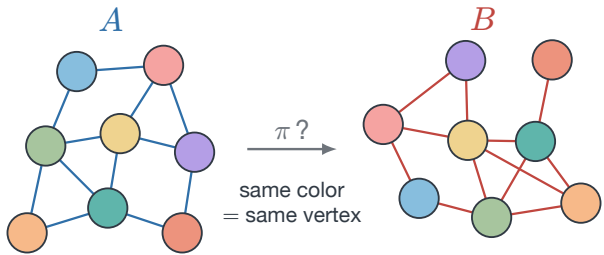
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Graph matching

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A natural formulation: relabel B to have as many edges in common with A as possible,

$$\max_{\pi \in \mathfrak{S}_n} \sum_{i<j} A_{ij} B_{\pi(i)\pi(j)}.$$

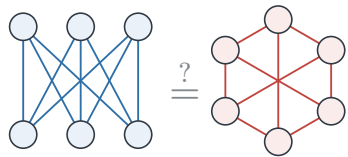
For the random models below, this is the maximum likelihood estimator.

Worst-case hardness

- $n!$ possible correspondences $(100! \approx 10^{158})$.
- Maximizing the overlap is a quadratic assignment problem Koopmans Beckmann 57. It is NP-hard, even to approximate Makarychev Manokaran Sviridenko 14

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- Even if A and B are identical up to relabeling (graph isomorphism), the best known algorithm is quasi-polynomial Babai 16



the same graph, relabeled?
(here: yes)

Average-case graph matching

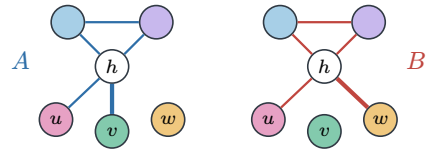
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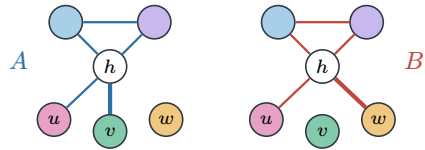


noise deleted hv and added hw
truth: 4 common edges
swap $v \leftrightarrow w$: 5 common edges

With weak correlation, wrong relabelings can fit better than the truth, so π cannot be recovered.

Average-case graph matching

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- Two questions. Statistical: can π be recovered at all? Computational: can it be recovered efficiently?
- Challenging even for simple random models. The most studied is the correlated Erdős–Rényi model Pedarsani Grossglauser 11. Although idealized, it took many years of work and shows a rich range of phenomena.

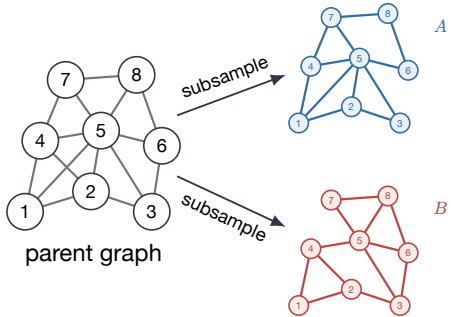


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Correlated Erdős–Rényi: Cullina Kiyavash 16, 17; Barak Chou Lei Schramm Sheng 19; Ganassali Massoulié 20; Mossel Xu 20; Ding Ma Wu Xu 21; Ganassali Massoulié Lelarge 21; Wu Xu Yu 22, 23; Fan Mao Wu Xu 23; Hall Massoulié 23; Mao Rudelson Tikhomirov 23; Mao Wu Xu Yu 23, 24; Ding Du 23; Ganassali Lelarge Massoulié 24; Ganassali Massoulié Semerjian 24; Ding Li 25; Ding Du Li 25; Huang Song Yang 25; Huang Yang 25, 26; …

The correlated Erdős–Rényi model



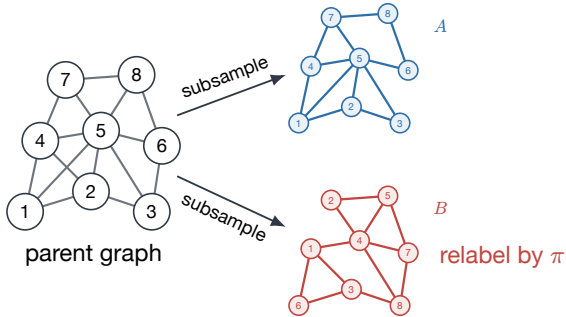
- $(A_{ij}, B_{\pi(i)\pi(j)})$ are i.i.d. pairs of Bern(q) with correlation r Pedarsani Grossglauser 11

In words:

r = how similar the two graphs are ($r = 1$: identical)

nq = average degree (sparse or dense)

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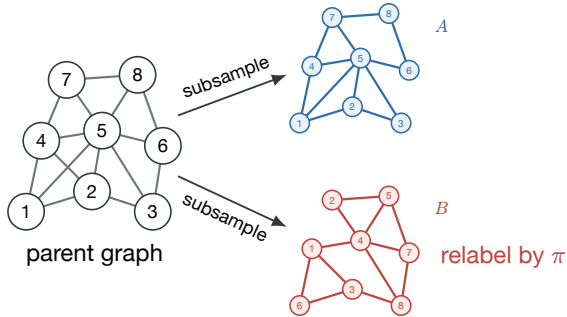
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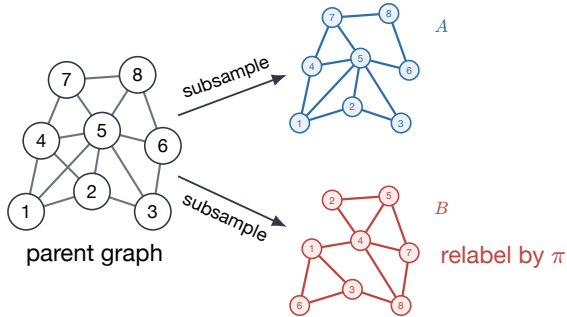
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- Goal: observe A and B; recover π as $n \to \infty$.

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Computational phenomena

polynomial time

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constant r close to 1		
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- Dense: degree-$O(1/r)$ polynomials fail Ding Du Li 25: a smooth tradeoff between correlation and running time

Beyond Erdős–Rényi

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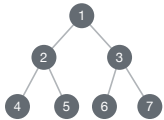
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inhomogeneous

Ding Fei Wang 25; Vassaux 26



growing graphs

Rácz Sridhar 22; Bäumler Rácz Ross Sridhar 26



block models

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geometric graphs

dependent edges: this talk

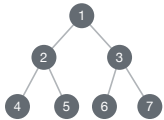
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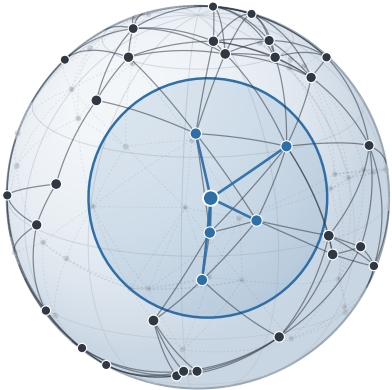
dependent edges: this talk

Which phenomena of correlated Erdős–Rényi persist in geometric graphs?
The dependence between edges makes the analysis much harder.

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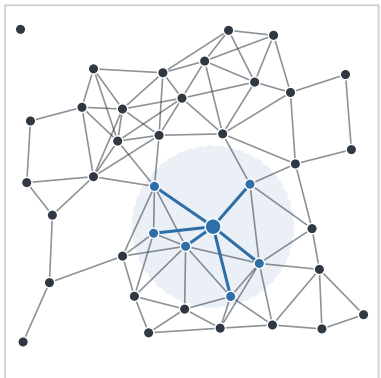
Random geometric graphs



on the sphere $\mathbb{S}^{d-1}$

- Latent vectors $u_1, \dots, u_n$ i.i.d. uniform on $\mathbb{S}^{d-1}$;
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  (u_j lies in a cap of measure p around u_i)

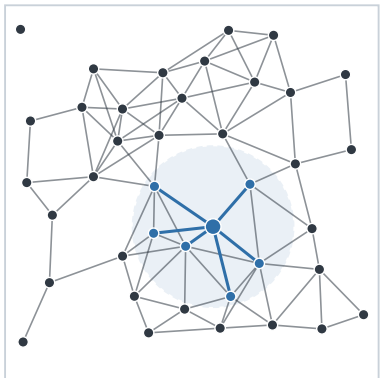
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in pictures: points in the plane

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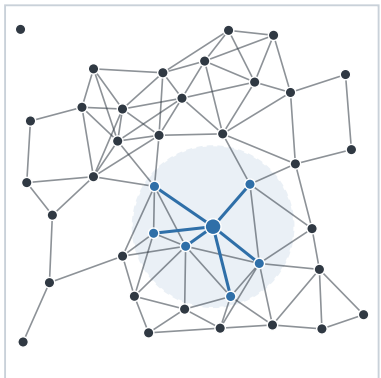
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- Well studied: testing for geometry Bubeck Ding Eldan Rácz 16; Liu Mohanty Schramm Yang 24; Du Mao Sun Wu Xu 26, expansion Liu Mohanty Schramm Yang 23, recovering the geometry L. Schramm 23; Fernandez Zhu 26. Matching: much less understood.

Correlated random geometric graphs

graph A



a vertex i
(and its partner in B)

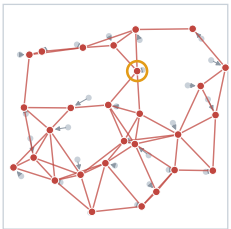
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perturb every latent point,
then connect nearby points

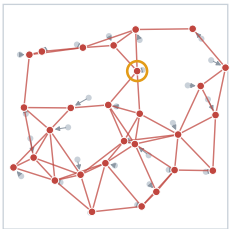
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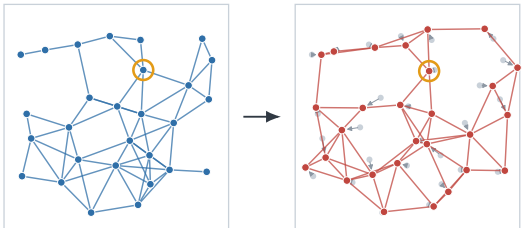
- u_i^A uniform on $\mathbb{S}^{d-1}$;
- u_i^B uniform on $\{v : \langle u_i^A, v\rangle = \rho\}$.

For a hidden permutation π, observe

$$A_{ij} = \mathbf{1}\{\langle u_i^A, u_j^A\rangle \geq \tau_{p,d}\},$$
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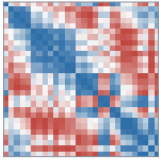
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- Known: n, d, p, ρ. Goal: exact recovery, $\mathbb{P}[\hat{\pi} = \pi] \to 1$. Dense: $p \in [p_0, 1 - p_0]$; $\rho \in (0, 1]$ fixed.

If we could see the latent vectors

- We would see two point clouds, but only up to a rotation (the inner products do not fix the coordinates). Find a rotation *and* a matching: Procrustes matching Wang Wu Xu Yolou 22; Gong Li 24; Even Ganassali Maier Massoulié 24; Niu Schramm Xu 26

inner products





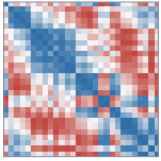
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Cartoon: Berliner Wespen, 1878 (public domain)

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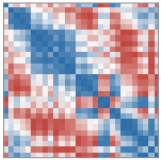
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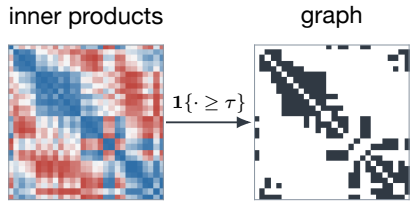
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In our model, we only observe whether each inner product exceeds a threshold.

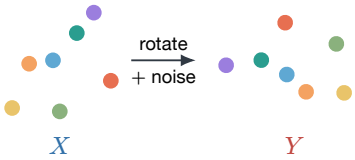
High-dimensional Procrustes matching

Niu Schramm Xu 26

Observe $X_1, \dots, X_n \sim N(0, I_d)$ and

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a rotated, noisy copy of the cloud, relabeled by π.



(same color = corresponding points; unknown in the data)

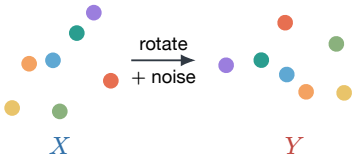
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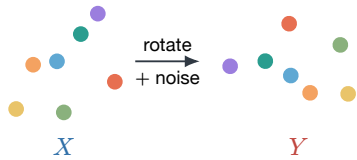
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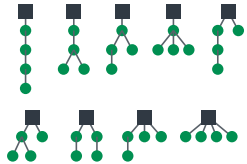
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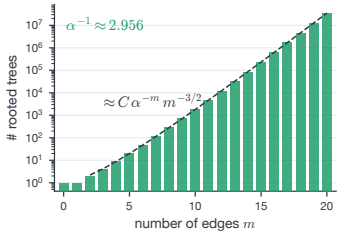
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Tree counting and Otter's constant

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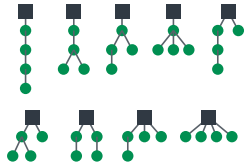


all 9 rooted trees with 4 edges

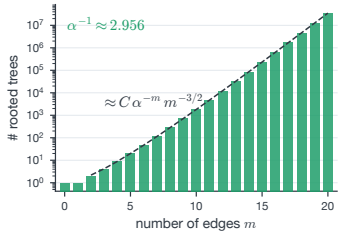


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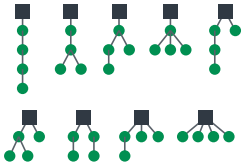


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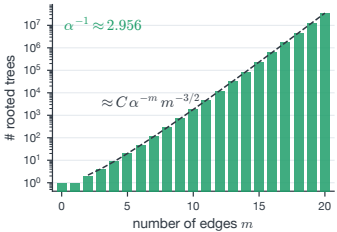
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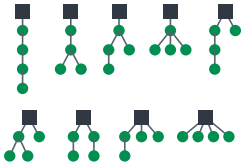


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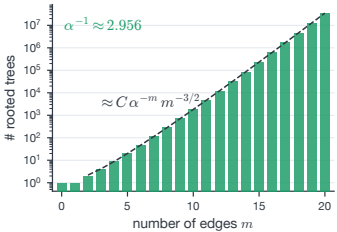
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- Correlated Erdős–Rényi: r is the edge correlation. Procrustes: effective $r = \rho^2$, so $\rho^2 > \sqrt{\alpha}$.



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Main result: efficient exact recovery

Theorem 1 Huang, Jiao, **L.**, Li 26

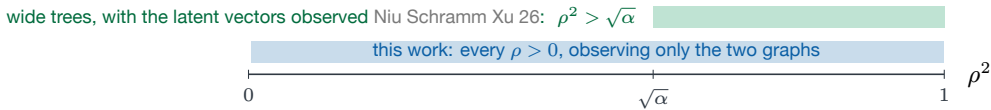
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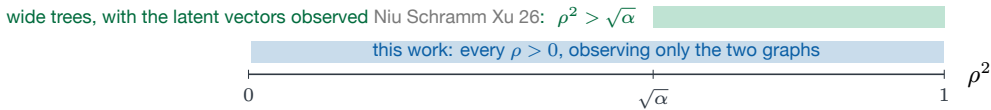


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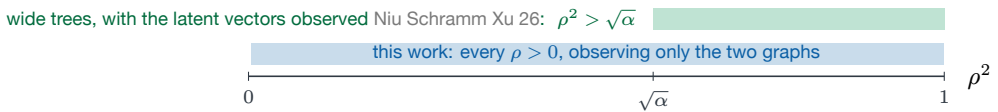


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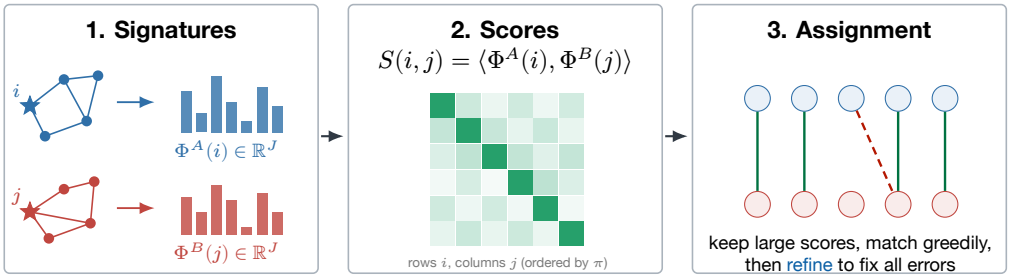
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- A robust framework, seeded path counts: the geometry enters only through spectral estimates and local comparisons with correlated Erdős–Rényi. We expect it to apply to other dense models.



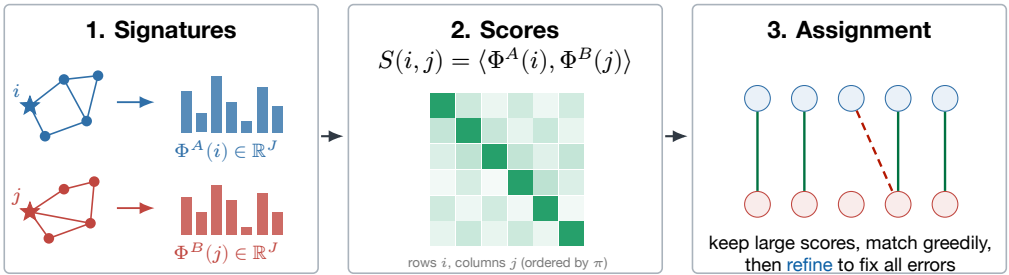
Outline

1. Graph matching: motivation and background
2. Correlated random geometric graphs and our result
3. **The algorithm: counting seeded paths**
4. Analysis: concentration and the latent geometry
5. Lower bounds and open problems

Signatures and similarity scores

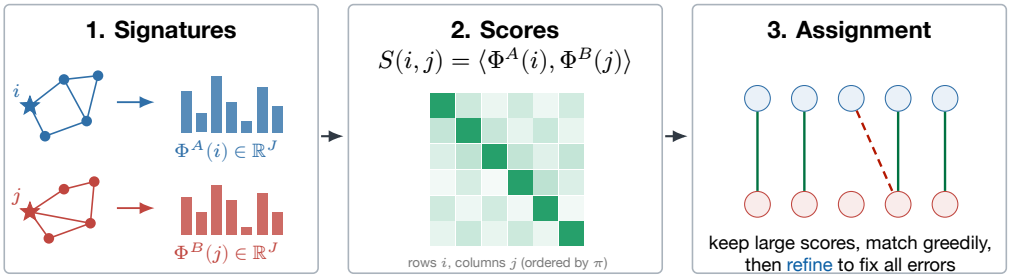


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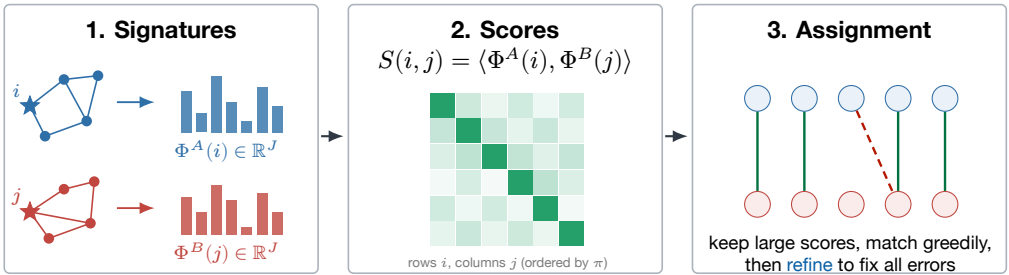
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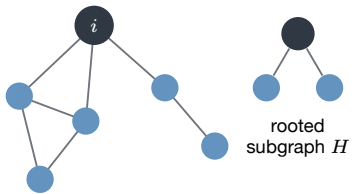
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How to construct the vertex signature?

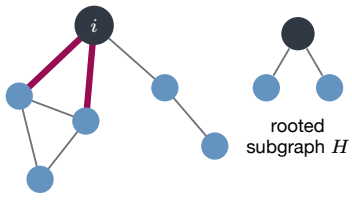
Rooted subgraph counts



graph A

- $W_{i,H}(A)$: number of copies of H rooted at i in A

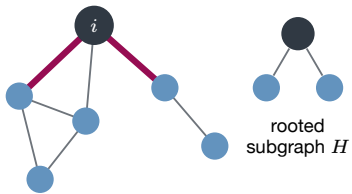
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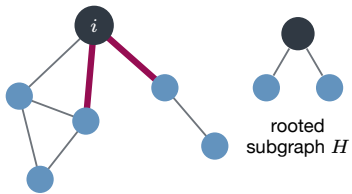
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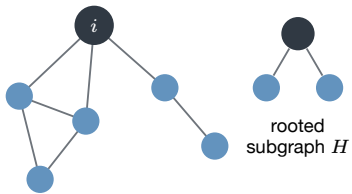
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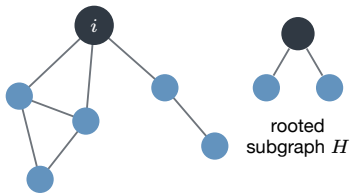


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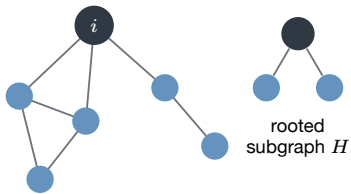
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- $W_{i,H}(A)$: number of copies of H rooted at i in A
- Signed counts, with centered entries $\overline{A} = A - \mathbb{E}A$ Bubeck Ding Eldan Rácz 16:

$$W_{i,H}(\overline{A}) = \sum_{\substack{S \cong H \\ \mathrm{root}(S)=i}} \prod_{e \in E(S)} \overline{A}_e$$

Rooted subgraph counts



graph A

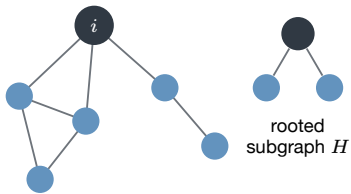
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- $W_{i,H}(A)$: number of copies of H rooted at i in A
- Signed counts, with centered entries $\overline{A} = A - \mathbb{E}A$ Bubeck Ding Eldan Rácz 16:

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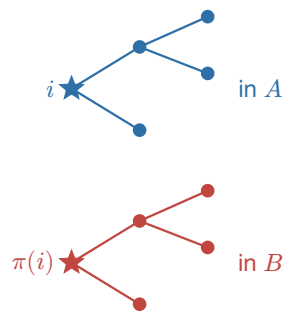
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- A rich family $\mathcal{H}$ of rooted graphs $\Rightarrow$ signature $\left(W_{i,H}(\overline{A})\right)_{H \in \mathcal{H}}$ Mao Wu Xu Yu 23

Similarity scores from signed subgraph counts

For a family $\mathcal{H}$ of rooted graphs with N edges Mao Wu Xu Yu 23:

$$\Phi_{ij} = \sum_{H \in \mathcal{H}} \mathrm{aut}(H)\, W_{i,H}(\overline{A})\, W_{j,H}(\overline{B}).$$

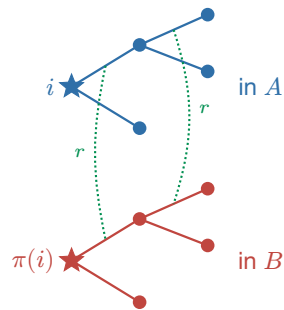


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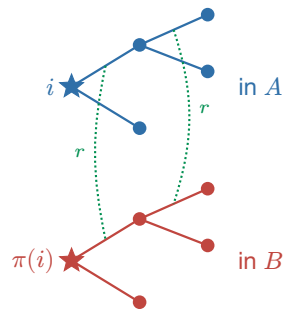


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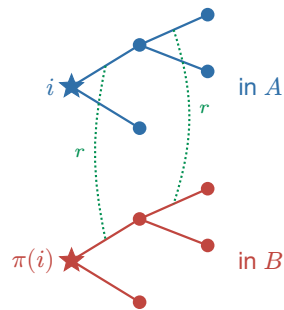
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- Different shapes are orthogonal under independence.

Heuristic: $\mathrm{SNR}^2 \approx \sum_{H \in \mathcal{H}} r^{2|E(H)|} = |\mathcal{H}|\, r^{2N}$



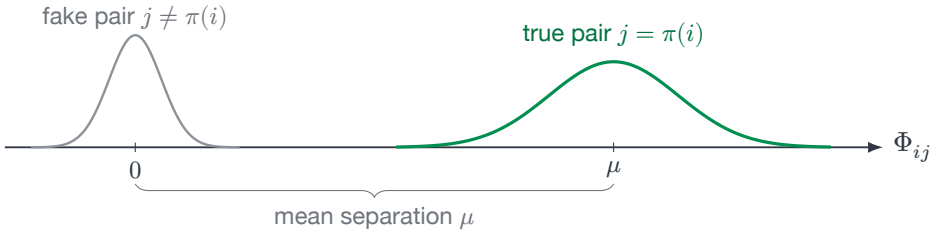
Desirable properties: mean and variance

For each vertex i in A and j in B: $\Phi_{ij} = \sum_{H \in \mathcal{H}} \mathrm{aut}(H)\, \underbrace{W_{i,H}(\overline{A})\, W_{j,H}(\overline{B})}_{\text{correlated iff } j=\pi(i)}$



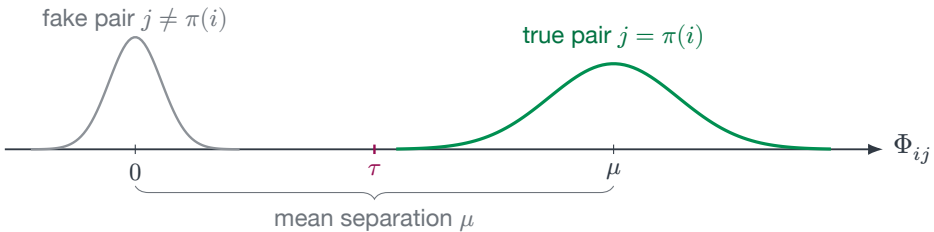
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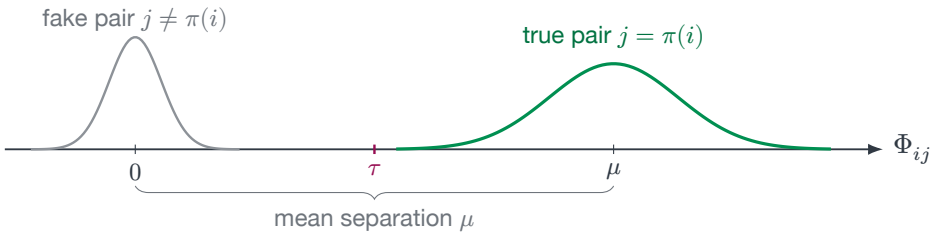
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Which family $\mathcal{H}$ of rooted graphs to count?

Counting trees

- Wishful thinking: ignore the correlations between different shapes H. Then

$$\frac{\mathrm{Var}[\Phi_{ij}]}{\mu^2} \approx \frac{1}{|\mathcal{H}|\, r^{2N}} \longrightarrow 0$$

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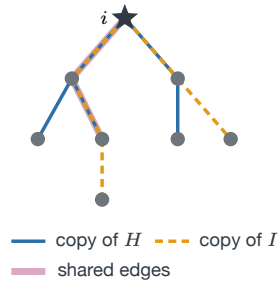
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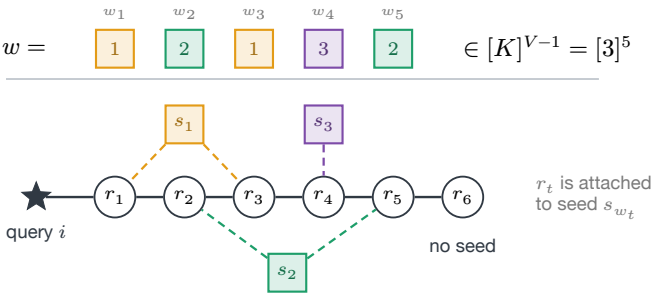
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- But copies of different trees can share edges: the cross-correlations cannot be ignored.



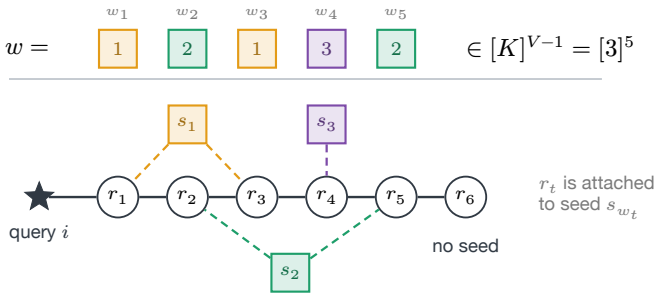
Our idea: seeds as labels

Challenge 1: only $\approx \alpha^{-m}$ tree shapes with m edges, so trees amplify only $r > \sqrt{\alpha}$.
Idea: suppose we know K seed pairs $(a, \pi(a))$, to be guessed later: labels readable in both graphs.



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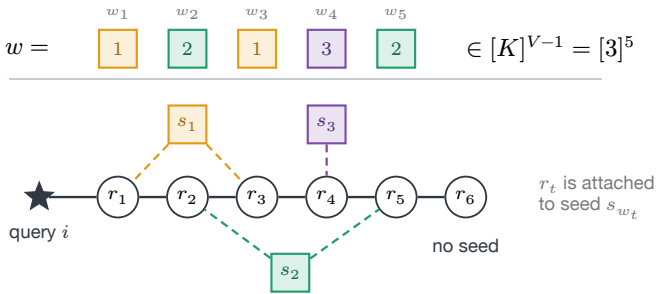
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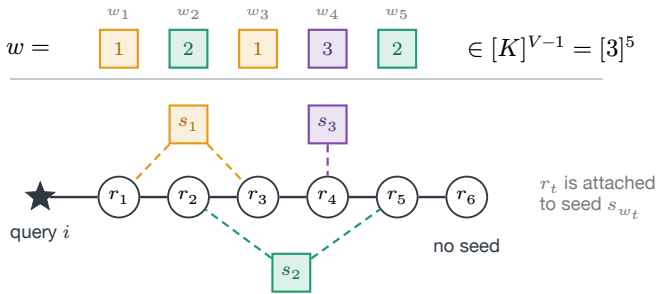
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- K^{V-1} words: K choices per step, for any K, the same in A and B.

A toy example



1. Guess $K = 100$ seed pairs, and pretend they are correct.

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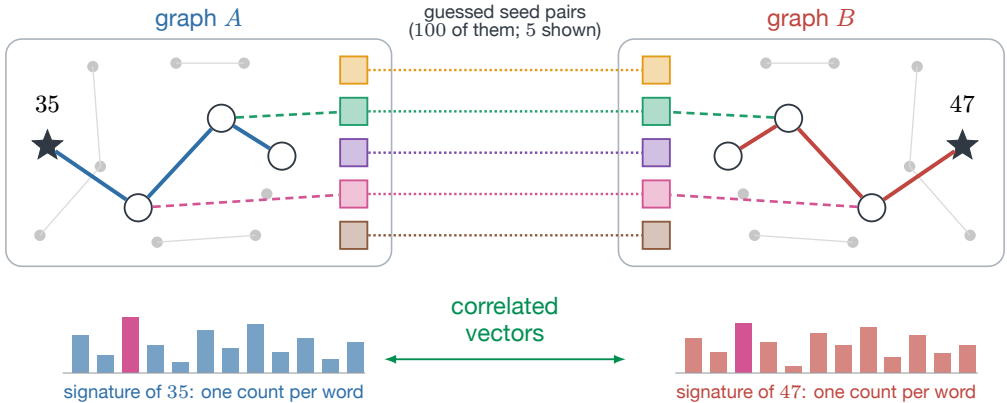
2. Pick a vertex, say 35 in A. Suppose it corresponds to 47 in B.

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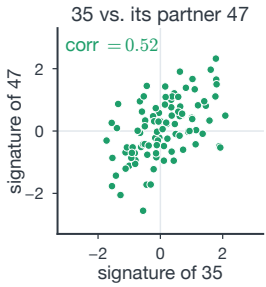
3. A word is a sequence of seeds, here **pink** then **green**: count the paths from 35 that pass next to these seeds, in order. Each has a twin from 47: the same people, mostly the same edges.

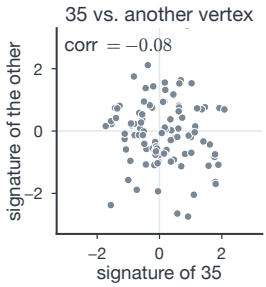
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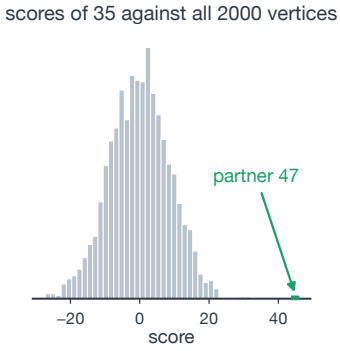


4. Do this for every word. The signature vectors of 35 and 47 are correlated; for a wrong vertex in B they are not.

With the right seeds, the partner stands out



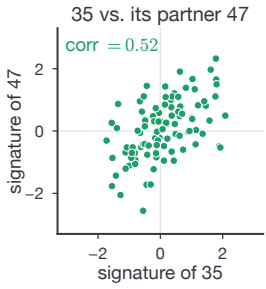


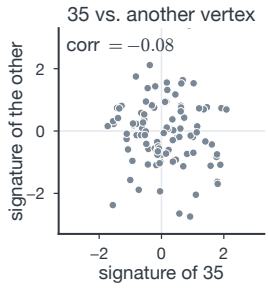


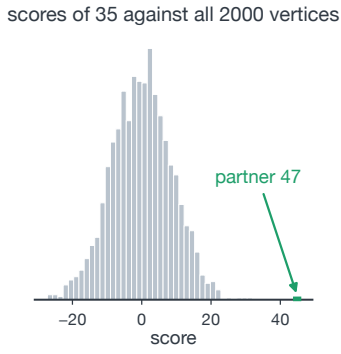
- Each dot is one seed. Signature of 35 against that of its partner 47: correlated; against another vertex: not.

Simulation: correlated random geometric graphs with $n = d = 2000$ (edge correlation ≈ 0.8) and $K = 100$ seeds; one coordinate per seed.

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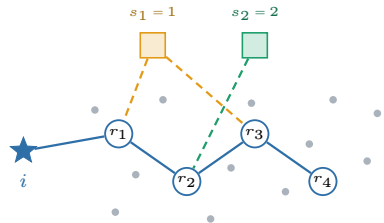
- Each dot is one seed. Signature of 35 against that of its partner 47: correlated; against another vertex: not.
- So the score $S(35, j) = \langle \Phi^A(35), \Phi^B(j) \rangle$ is largest at the partner $j = 47$.

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Seeded path signatures (high-dimensional regime)

- Centered entries $Z^G_{uv} = \frac{G_{uv} - p}{\sqrt{p(1-p)}},\ G \in \{A, B\}$
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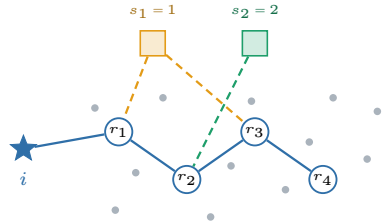


one summand for $w = (1, 2, 1)$ in graph A

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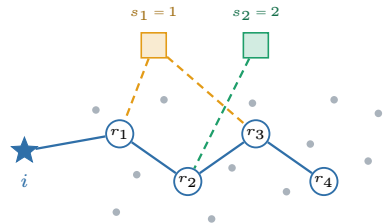
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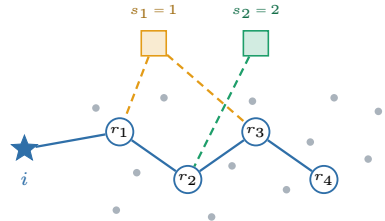
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Amplification: each word has $2V - 1$ edges, so its squared signal is $\approx r^{2(2V-1)}$. Summing over K^{V-1} words, $\text{SNR}^2 \approx r^2 (K r^4)^{V-1}$, which grows once $K > r^{-4}$: any fixed $r > 0$.

Polynomial time: color coding

Alon Yuster Zwick 95

Exhaustive search over paths: $n^{\Theta(V)}$ time.

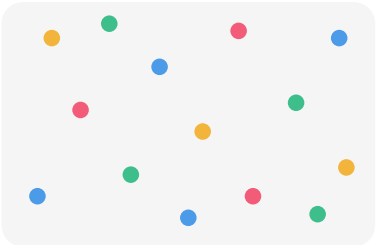
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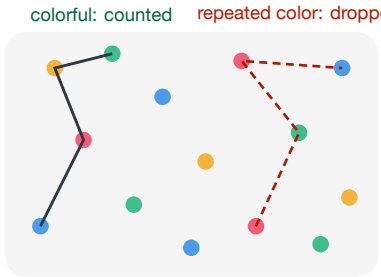
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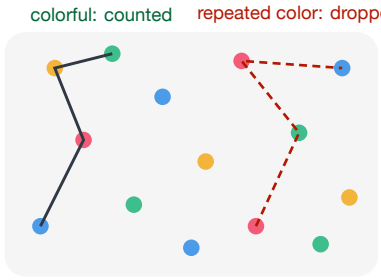
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Polynomial time: color coding

Alon Yuster Zwick 95

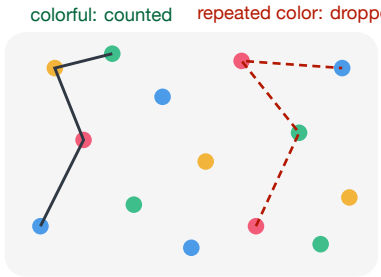
Exhaustive search over paths: $n^{\Theta(V)}$ time.

1. Color each vertex uniformly from $[V]$.
2. Count colorful paths (distinct colors) by dynamic programming over (level, used colors, vertex):

$$F_t(Q, r) = Z_{s_{w_t} r} \sum_q Z_{rq} F_{t+1}(Q \setminus \{\chi(r)\}, q).$$

3. Rescale by $p_{\text{col}} = \frac{V!}{V^V} \approx e^{-V}$ and average over M colorings.

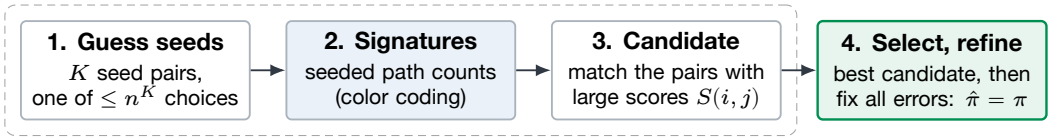
$V = \Theta(\log \log n)$: e^{-V} and 2^V are polylog(n). Time $n^{2+o(1)}$ per seed guess; total $O(n^{K+2+o(1)} + n^3)$.



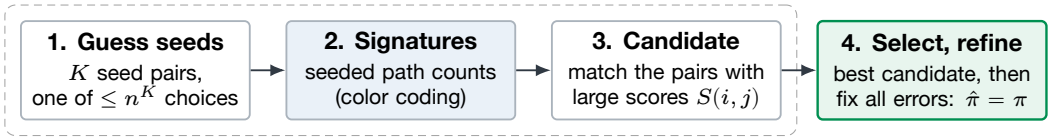
$V = 4$ colors

Color coding for subgraph counts in statistics: Hopkins Steurer 17; Mao Wu Xu Yu 23, 24; Li 25

The algorithm

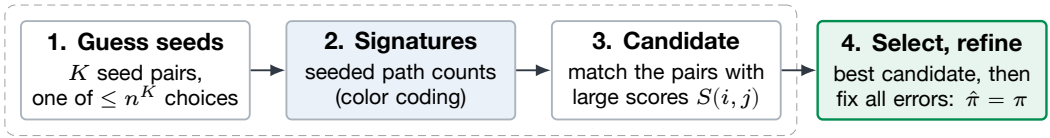


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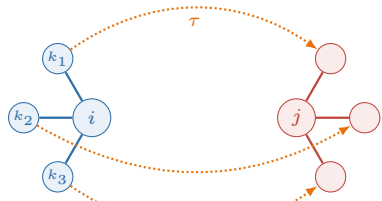


- Select the candidate τ with the largest overlap $\sum_{i<j} \overline{A}_{ij}\, \overline{B}_{\tau(i)\tau(j)}$.

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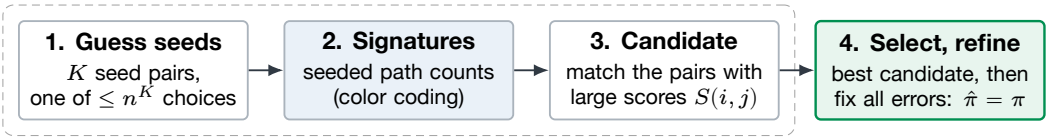


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- Refine: $\hat{\pi}(i) = \arg\max_j \sum_k \overline{A}_{ik}\, \overline{B}_{j\tau(k)}$.



refine: neighbors of i map to neighbors of $j \Rightarrow j = \pi(i)$

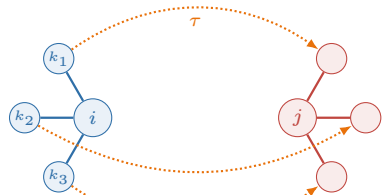
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Proposition

W.h.p., *simultaneously* for all lists of $\leq n^K$ candidates containing one with $\leq \varepsilon n$ errors, the output is π.

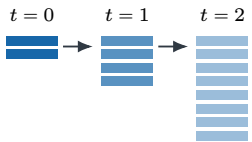


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Connection to the iterative algorithm of Ding–Li

Ding–Li Ding Li 25ab: guess K_0 seeds, then iterate:

- aggregate features through the adjacency matrix; apply nonlinear maps;
- more features, each weaker: $K_{t+1} \gg K_t$, $\varepsilon_{t+1} < \varepsilon_t$, but $K_t \varepsilon_t^2 \uparrow$.

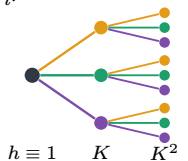


more features, each weaker

Ours: the same amplification as path counts. For each seed a,

$$h \longmapsto \left(Z_{s_a r} \sum_{r'} Z_{r r'}\, h(r') \right)_r .$$

Start at $h \equiv 1$, apply $w_{V-1}, \ldots, w_1$, contract with $Z_{i\cdot}$.



each step: one path edge + one seed edge

Linearized BP $\rightsquigarrow$ non-backtracking walks Krzakala et al. 13; Bordenave Lelarge Massoulié 15; Mossel Neeman Sly 18. Similarly, word counts give this amplification with explicit polynomials, on self-avoiding paths.

Outline

1. Graph matching: motivation and background
2. Correlated random geometric graphs and our result
3. The algorithm: counting seeded paths
4. **Analysis: concentration and the latent geometry**
5. Lower bounds and open problems

Concentration: paths of length $\log \log n$ suffice

- Chebyshev + union bound over n^2 false pairs needs $\mathrm{SNR}^2 \geq n^{2+\varepsilon}$: templates of size $\Theta(\log n)$

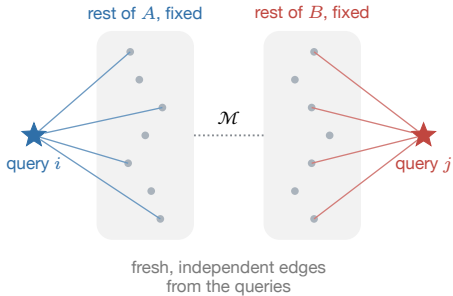
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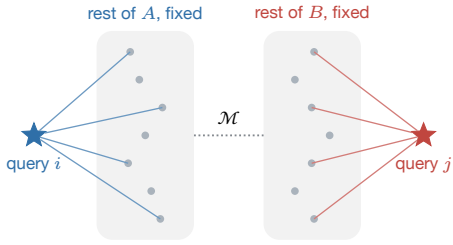
- Ours: $S(i,j) = (y_i^A)^\top \mathcal{M}\, y_j^B$, where $y_k^G = (Z_{kr}^G)_r$ are the edges at k and $\mathcal{M}$ counts the rest of the paths. Given the rest, for $j \neq \pi(i)$: a bilinear form in two (nearly) independent vectors.



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- Hanson–Wright type tail Hanson Wright 71; Hsu Kakade Zhang 12:

$$\mathbb{P}\big[|S(i,j)| \geq t \,\big|\, \text{rest}\big] \leq Ce^{-c\min\{t^2/(\kappa^2 J),\ t/\kappa\}}.$$



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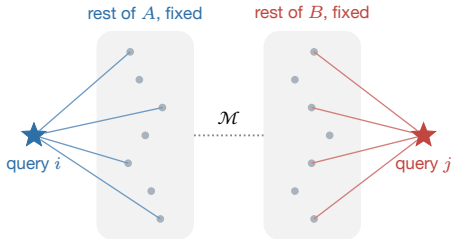
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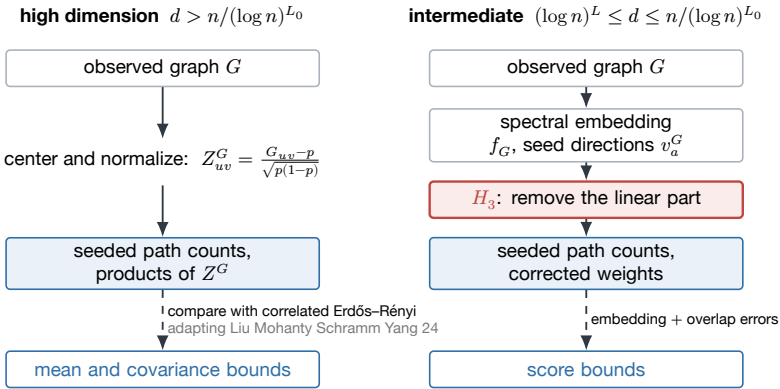
$$\mathbb{P}\big[|S(i,j)| \geq t \,\big|\, \text{rest}\big] \leq Ce^{-c\min\{t^2/(\kappa^2 J),\ t/\kappa\}}.$$

- So $\mathrm{SNR}^2 = \mathrm{polylog}(n)$ already beats all n^2 false pairs.



Geometric dependence: two regimes

Challenge 2: geometric correlations are small, but they accumulate over many path embeddings.



The split follows when the latent geometry can be recovered spectrally: Araya De Castro 19; Eldan Mikulincer Pieters 22; **L.** Schramm 23; Mao Zhang 24; Fernandez Zhu 26

Intermediate dimension: use the geometry

- In intermediate dimension, we first make use of the geometric information.

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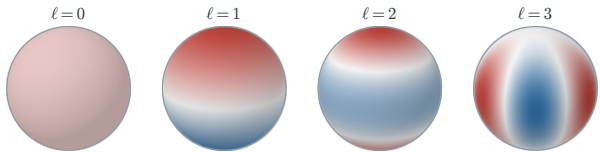
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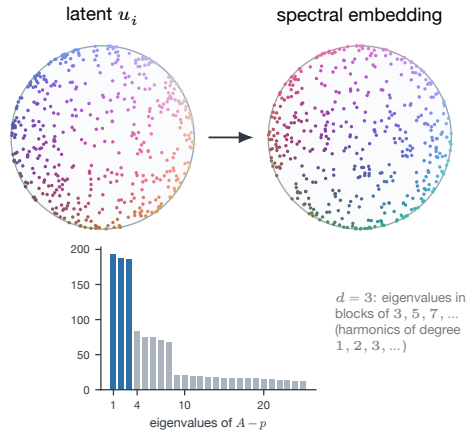
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- Step 2: count seeded paths with the estimated latent vectors instead of the raw edges. They are more accurate.
- Along the way, spherical harmonics show up (the Fourier modes of the sphere) Dai Xu 13.



spherical harmonics of degree $0, 1, 2, 3$

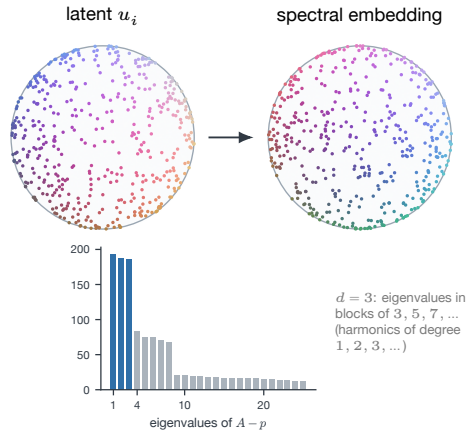
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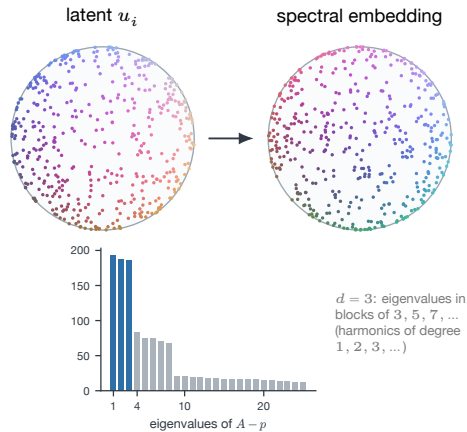
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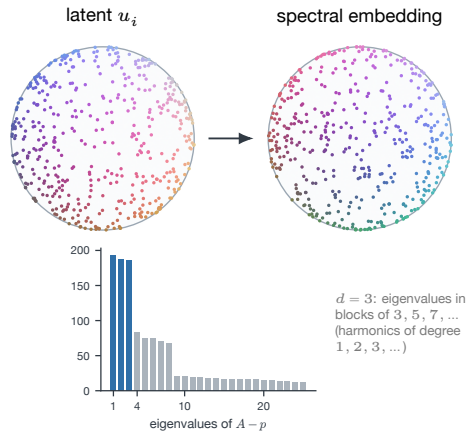


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- Scalar contractions are rotation invariant: the two embeddings may use unrelated coordinates.

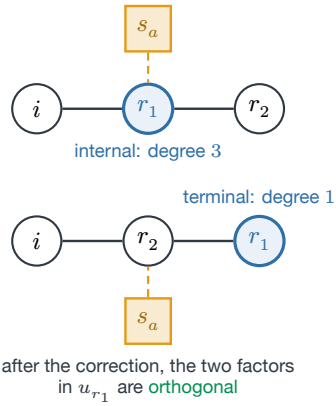


The harmonic correction H_3

- As a function of $z \in \mathbb{S}^{d-1}$, the cubic $(z^\top v)\, zz^\top$ has spherical harmonics of degree 3 and 1, just as $x^3 = \mathrm{He}_3(x) + 3x$.

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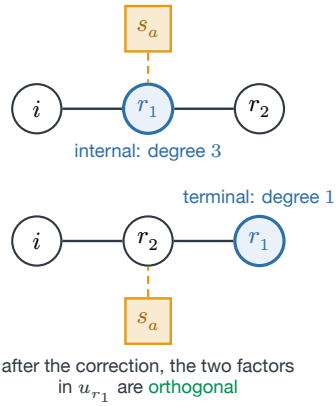
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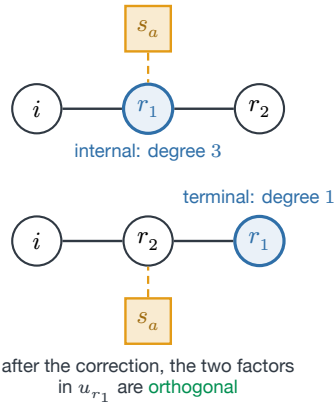
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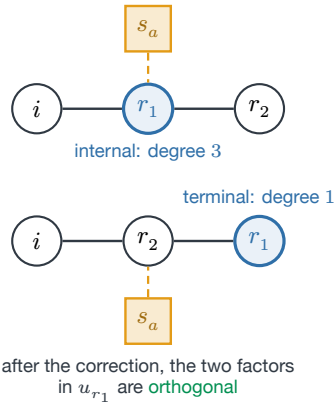
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Summary: chandeliers vs. seeded paths

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templates	unlabeled trees	trees in the point–coordinate graph	paths + K seed labels
growth of family	α^{-1} per edge	α^{-1} per edge pair	$\sqrt{K}$ per edge, any K
threshold	$r > \sqrt{\alpha}$	$\rho^2 > \sqrt{\alpha}$	any $r > 0$
template size	$\Theta(\log n)$	$\Theta(\log n)$	$\Theta(\log \log n)$
seeds	none	none	K guessed: n^K tries
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The price of the seeds: time $n^{K+2+o(1)}$ with $K \gtrsim r^{-4}$, so the exponent grows as $r \downarrow 0$ (consistent with the low-degree lower bound, at the end).

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Low-degree lower bound

Theorem 2 Huang, Jiao, **L.**, Li 26

$\mathbb{Q}$: two *independent* geometric graphs; $\mathbb{P}$: the correlated model, π uniform. If $D\rho^2 \to 0$ and $D/d \to 0$, then

$$\sup_{\deg f \le D} \frac{\left|\mathbb{E}_{\mathbb{P}} f - \mathbb{E}_{\mathbb{Q}} f\right|}{\sqrt{\mathrm{Var}_{\mathbb{Q}} f}} = o(1).$$

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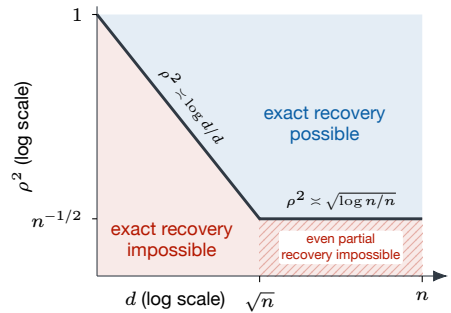
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- Idea: reduce to the χ^2-divergence between the Gram matrices of D aligned pairs and of two independent clouds inspired by Li 26; a harmonic expansion gives $\chi^2 \lesssim D^2(\rho^2 + d^{-1})^2$.

Information-theoretic limits

companion paper II

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$$\rho^2 \asymp \max\left\{\frac{\log d}{d}, \sqrt{\frac{\log n}{n}}\right\}.$$



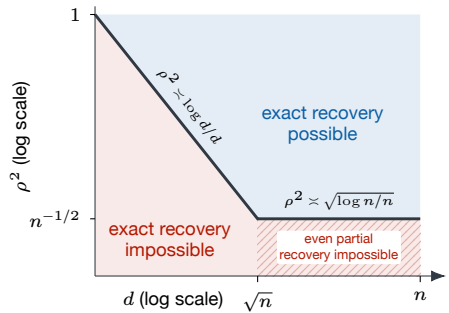
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- $\sqrt{\log n/n}$: from comparing edges, as for Gaussian edge weights with correlation ρ^2.



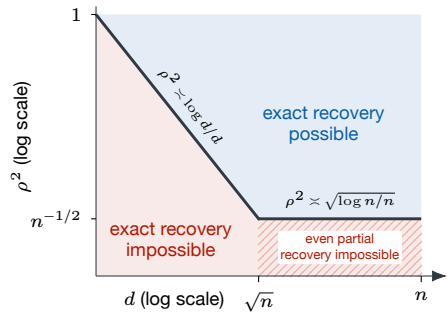
Information-theoretic limits

companion paper II

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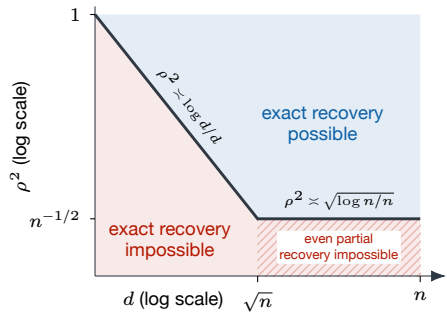
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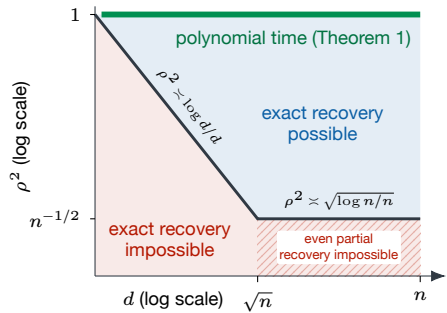
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- A vanishing threshold, against constant correlation for efficient algorithms: an information–computation gap.



Summary



Result. Exact recovery in polynomial time for correlated random geometric graphs, at every fixed correlation $\rho > 0$ (for $d \geq (\log n)^L$), beyond the $\sqrt{\alpha}$ barrier of tree counting.

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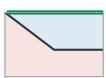
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Low-degree and information-theoretic results. Low-degree tests need degree $\gtrsim \rho^{-2}$, a smooth tradeoff; the information-theoretic threshold is $\rho^2 \asymp \max\{\frac{\log d}{d}, \sqrt{\frac{\log n}{n}}\}$.

Future directions

- **General kernels and sparse geometric graphs.** Replace the threshold by a general connection function: when do seeded path statistics still amplify the signal? Let $p \to 0$, ultimately $p = n^{-1+o(1)}$: does the absence of a correlation threshold persist in sparse regimes?

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Thank you very much!

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Images

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Procrustes cartoon: *Berliner Wespen* 1878 (public domain, Wikimedia Commons)